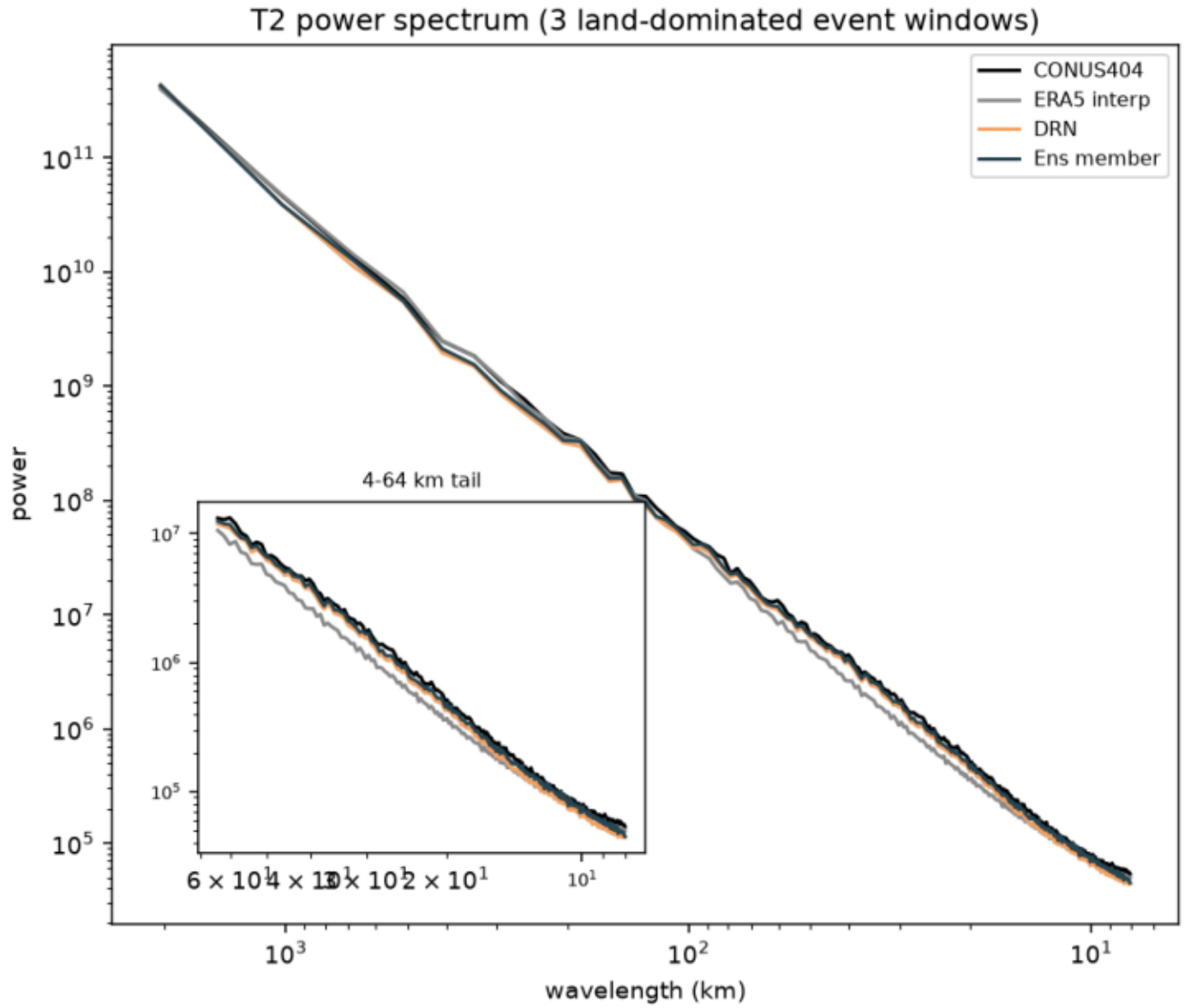
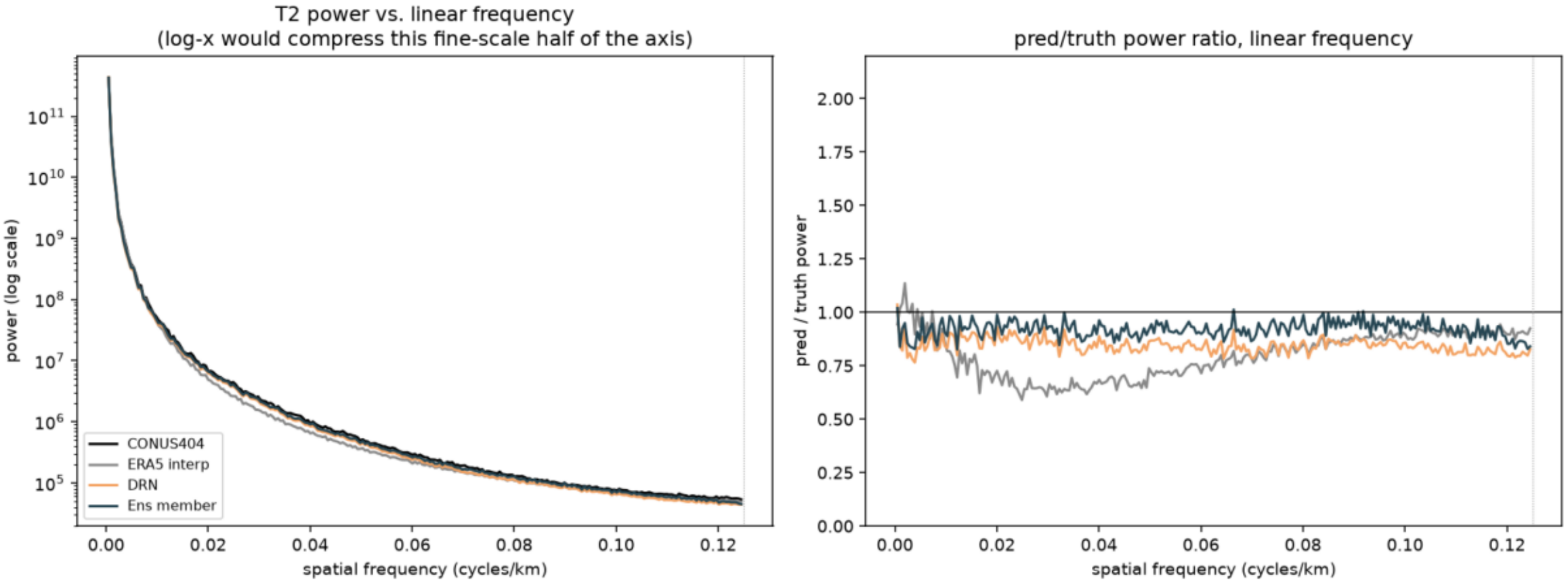
ERA5→CONUS404 Latent CorrDiff

Frequency-domain detail · NVIDIA CorrDiff comparison · Anderson Creek Fire case study

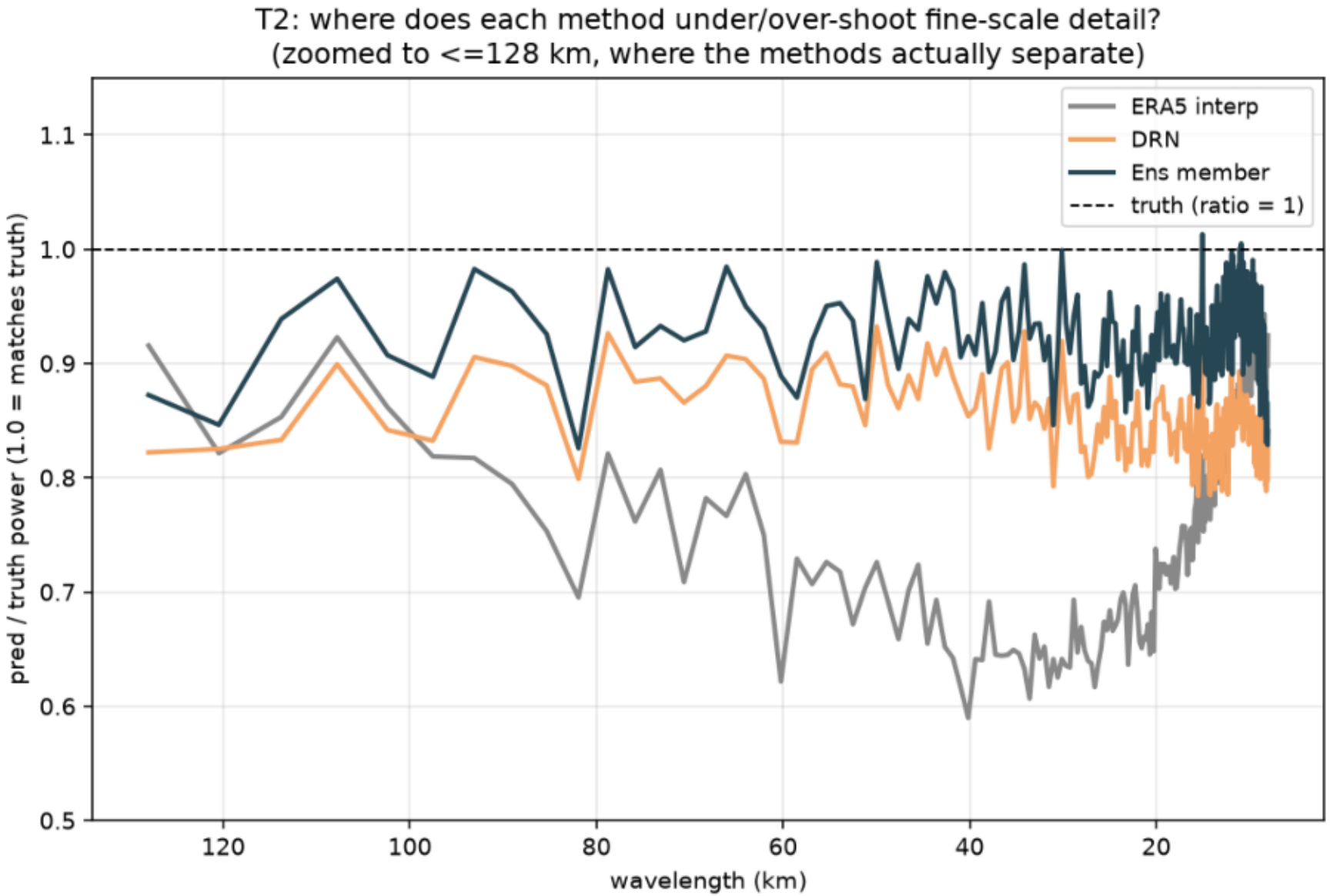
T2 power spectrum (log-log) — raw view, all methods look nearly identical



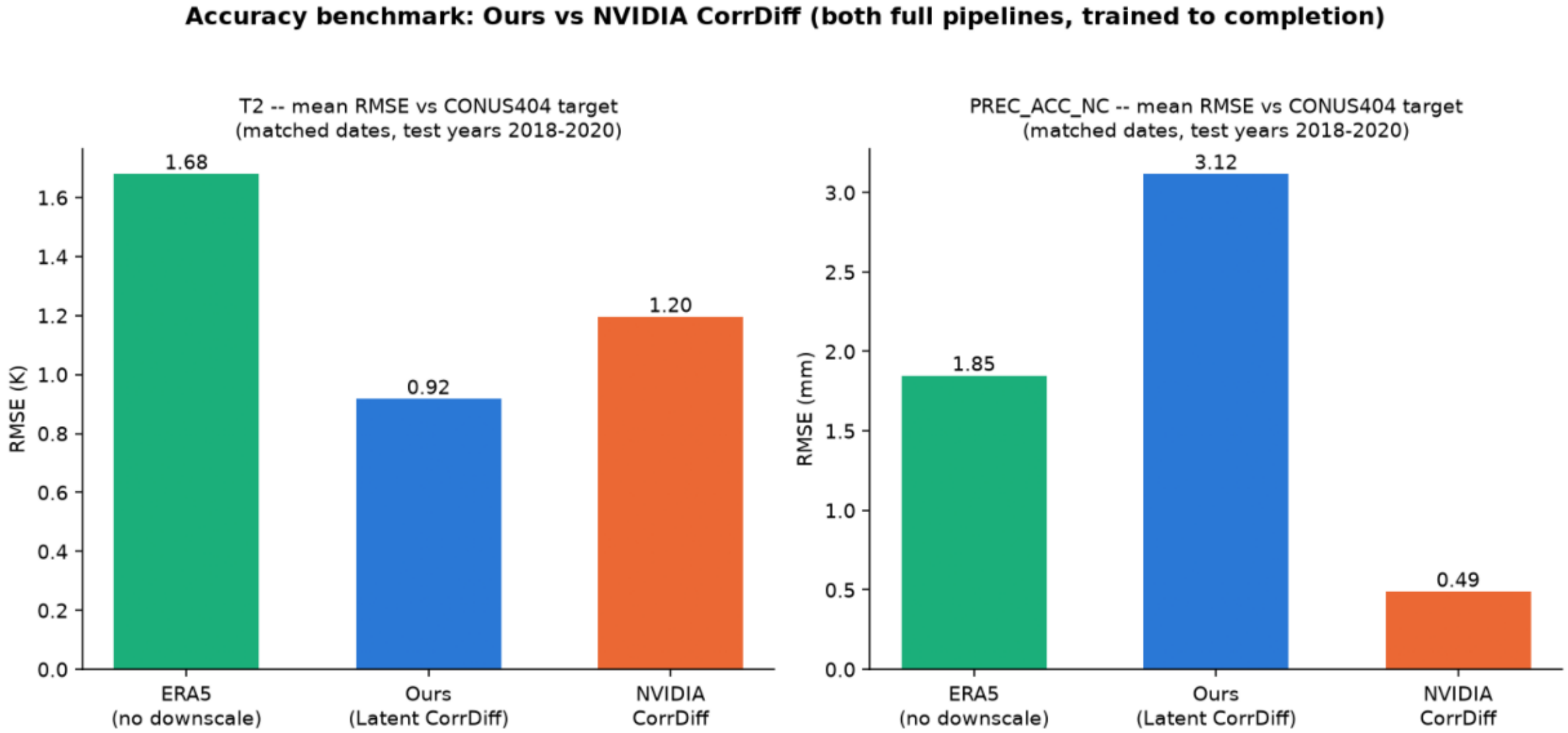
Same data, linear frequency axis — right panel = pred/truth power ratio



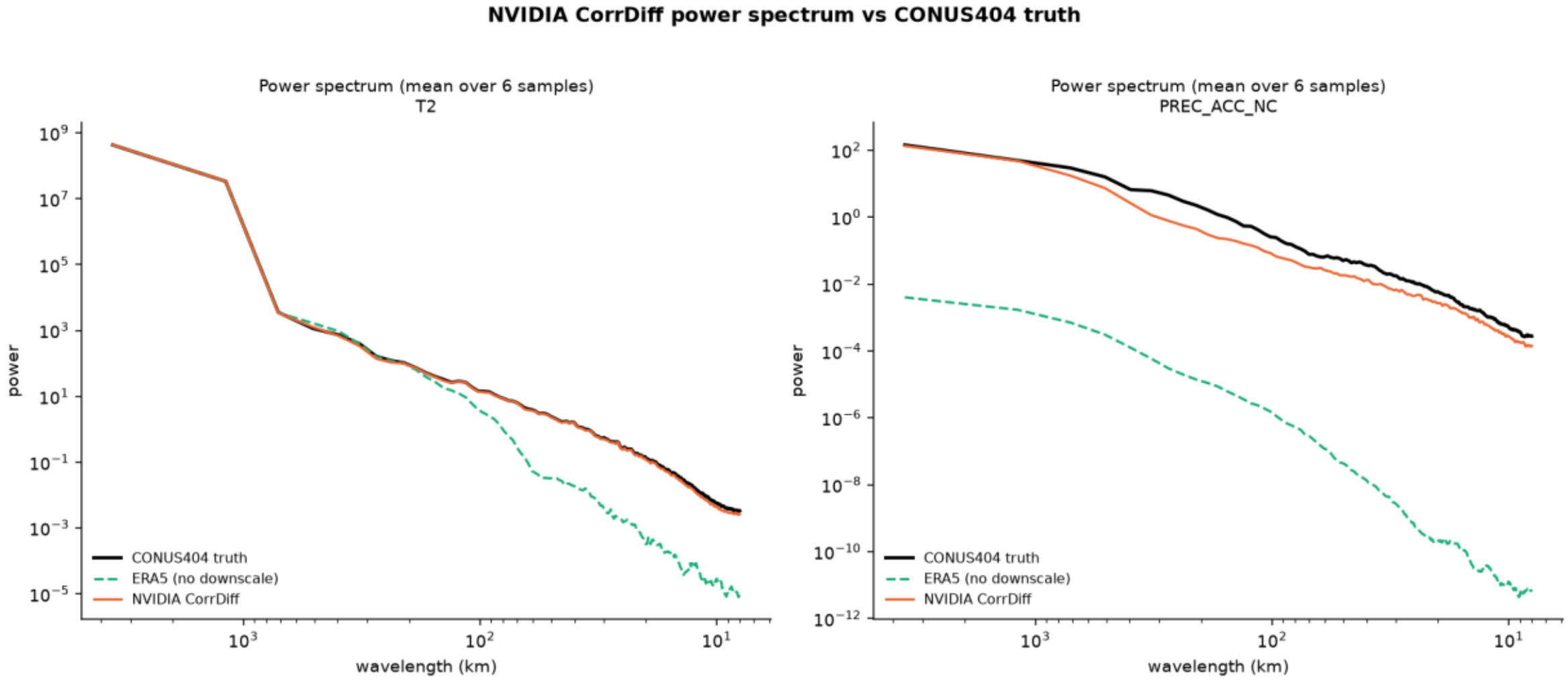
Zoomed pred/truth ratio, 4–128 km — where the methods actually separate

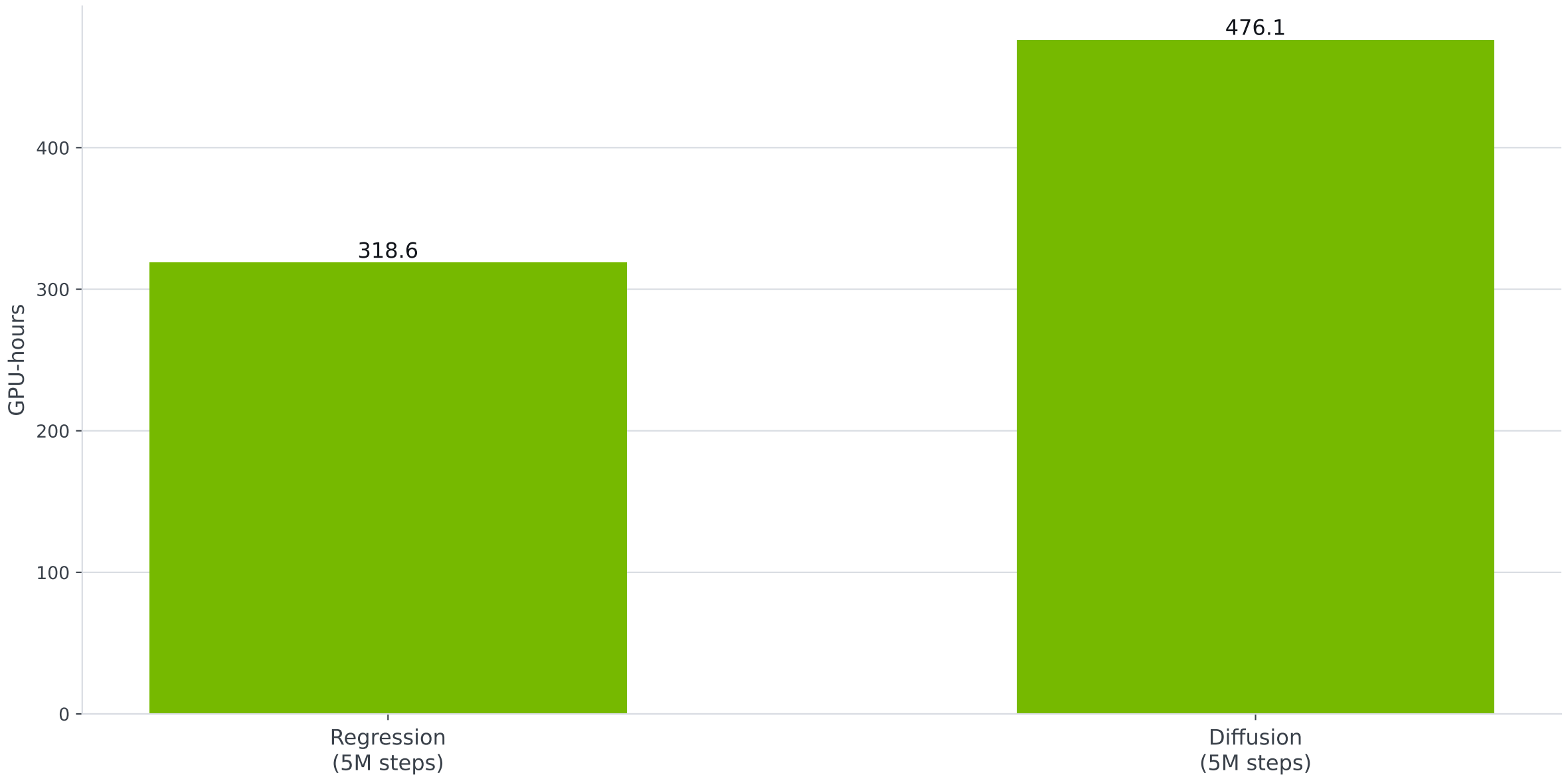


Ours vs. NVIDIA CorrDiff — mean RMSE, 6 matched held-out dates

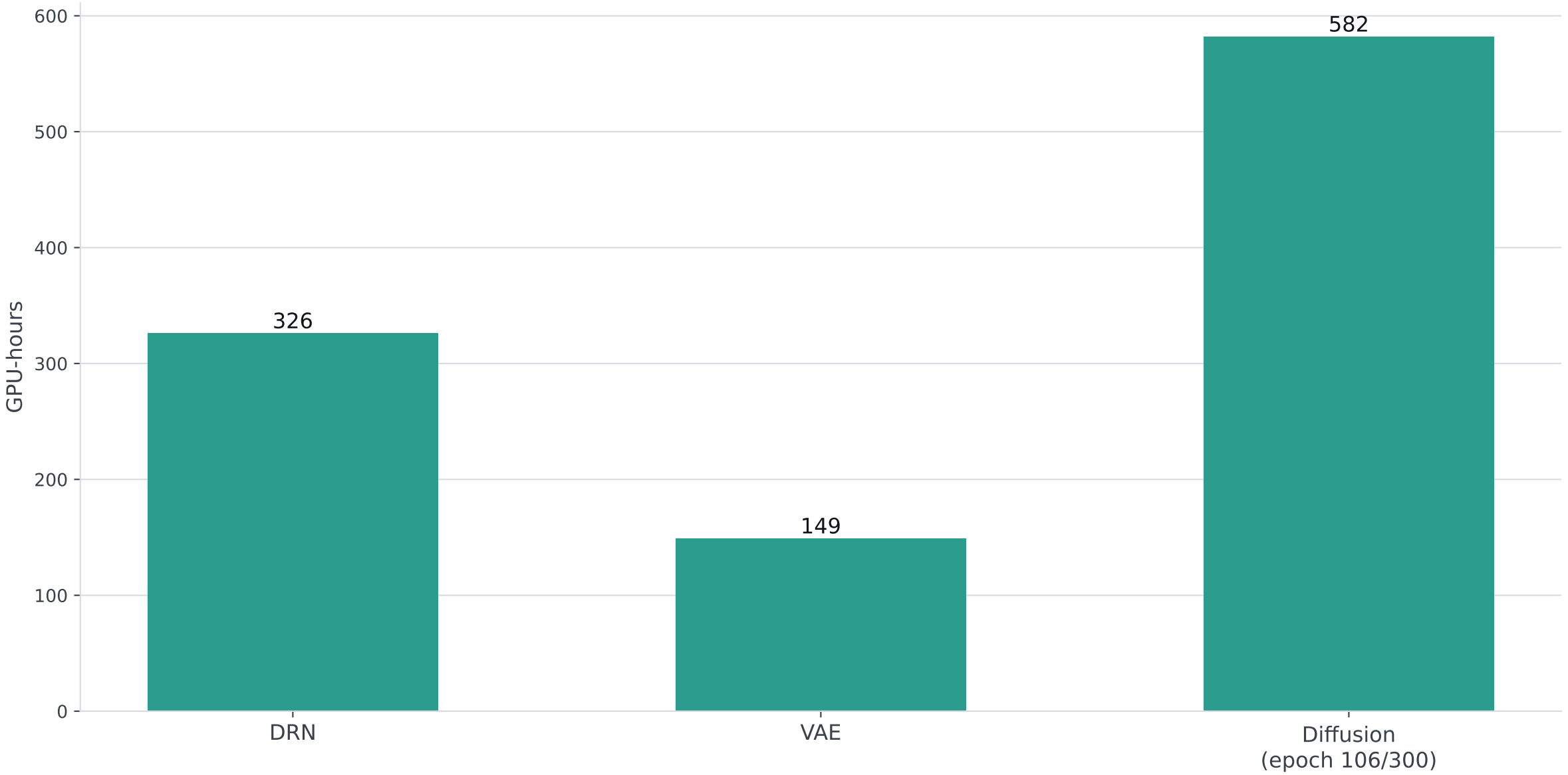


NVIDIA CorrDiff power spectrum, same 6 dates

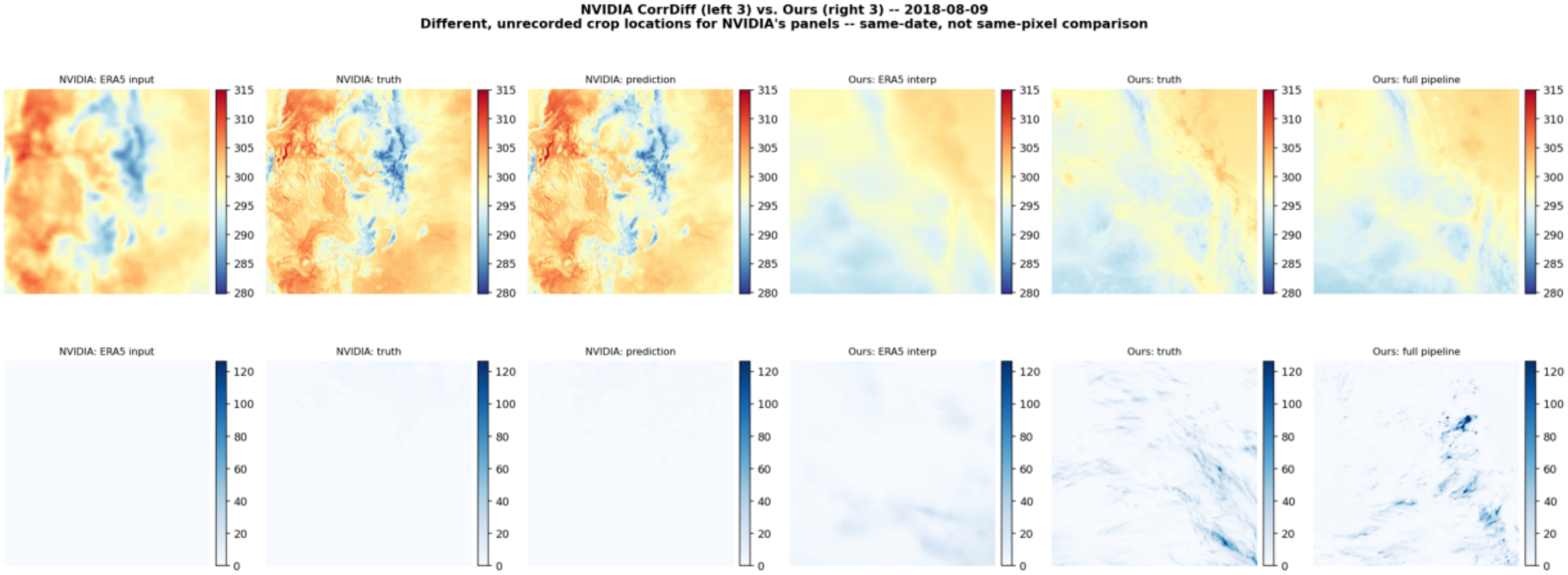



NVIDIA CorrDiff (official physicsnemo code) — training time, one clean pass
GPU-hours
0
100
200
300
400
318.6
476.1
Regression
(5M steps)
Diffusion
(5M steps)

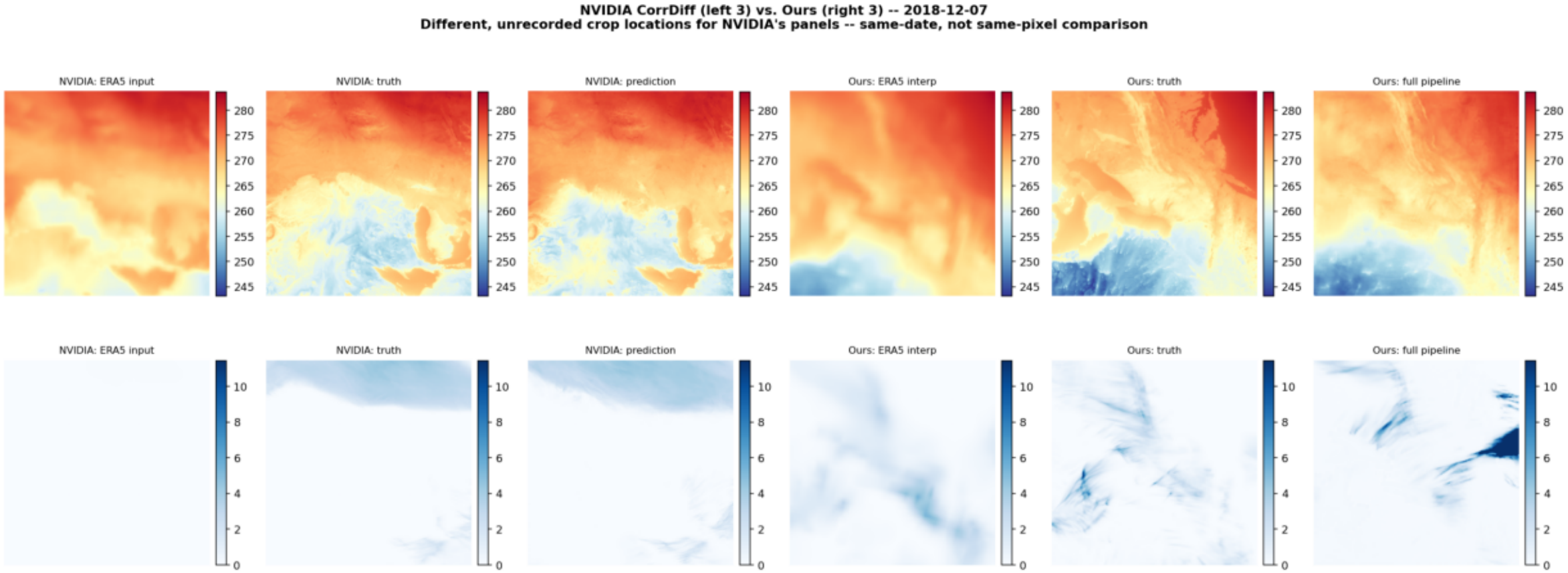
Our latent model (DRN + VAE + Diffusion) — training time, clean path only



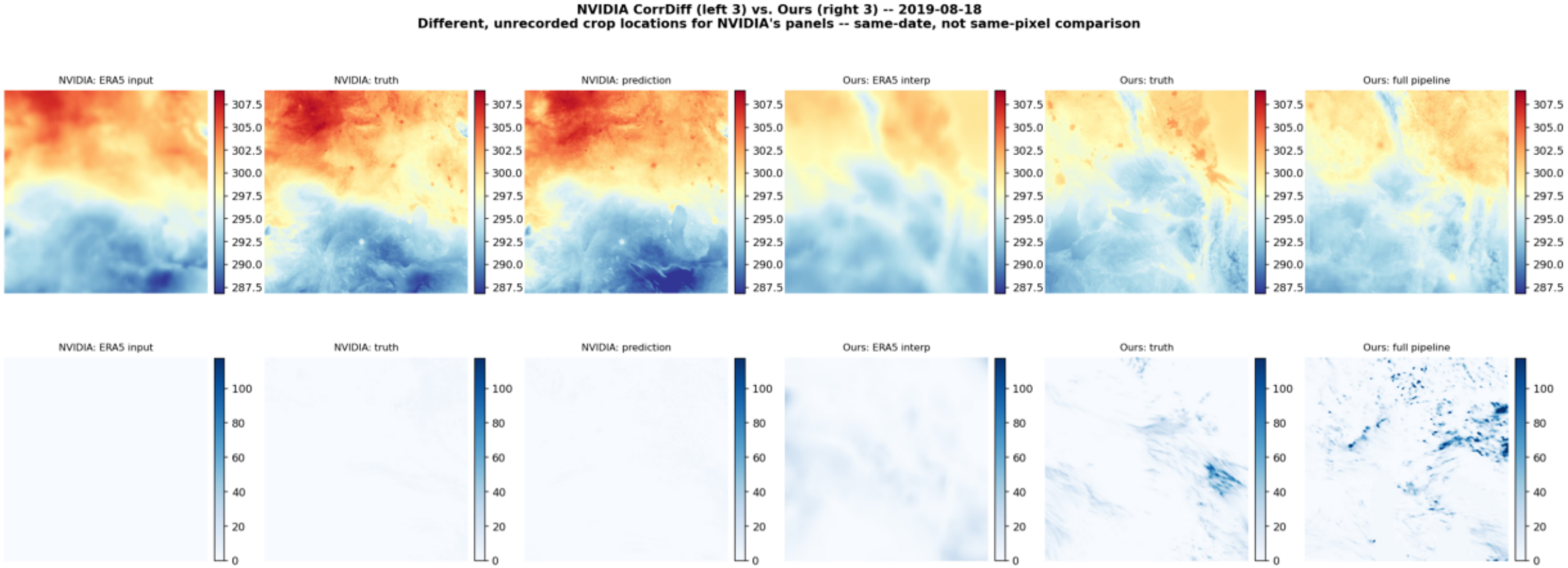
NVIDIA CorrDiff vs. ours, 2018-08-09 — same date, different (unrecorded) crops



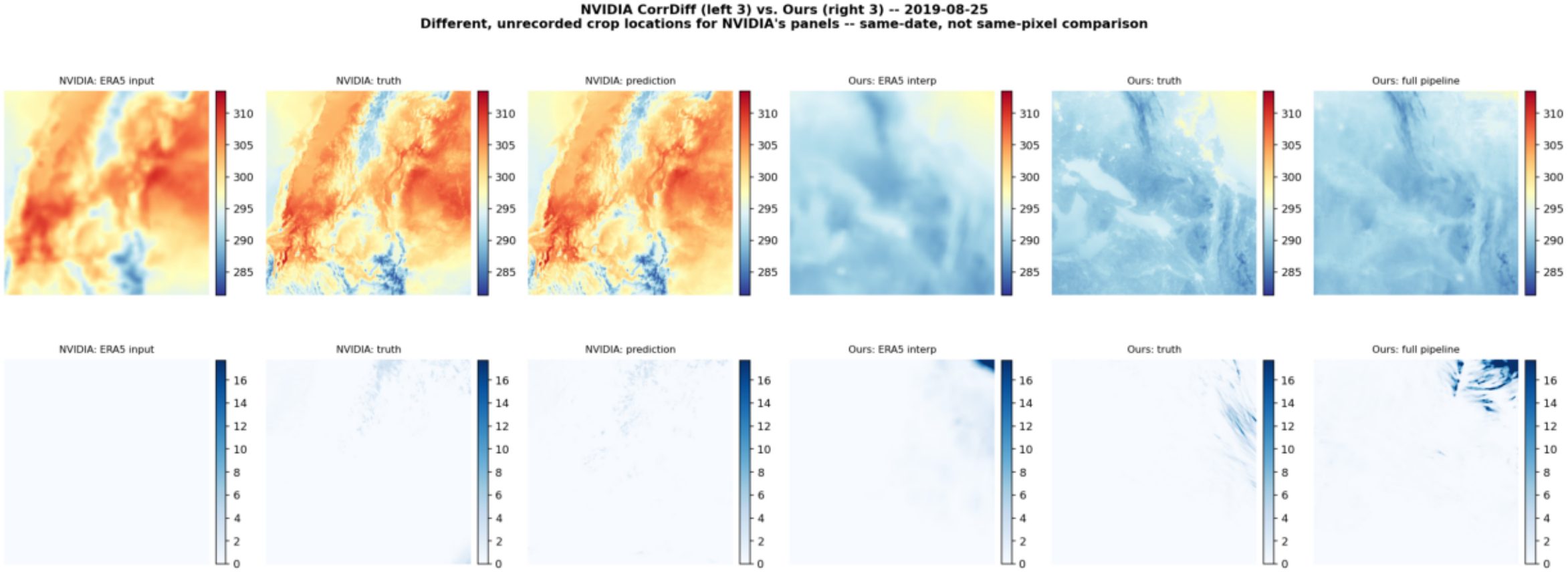
NVIDIA CorrDiff vs. ours, 2018-12-07 — same date, different (unrecorded) crops



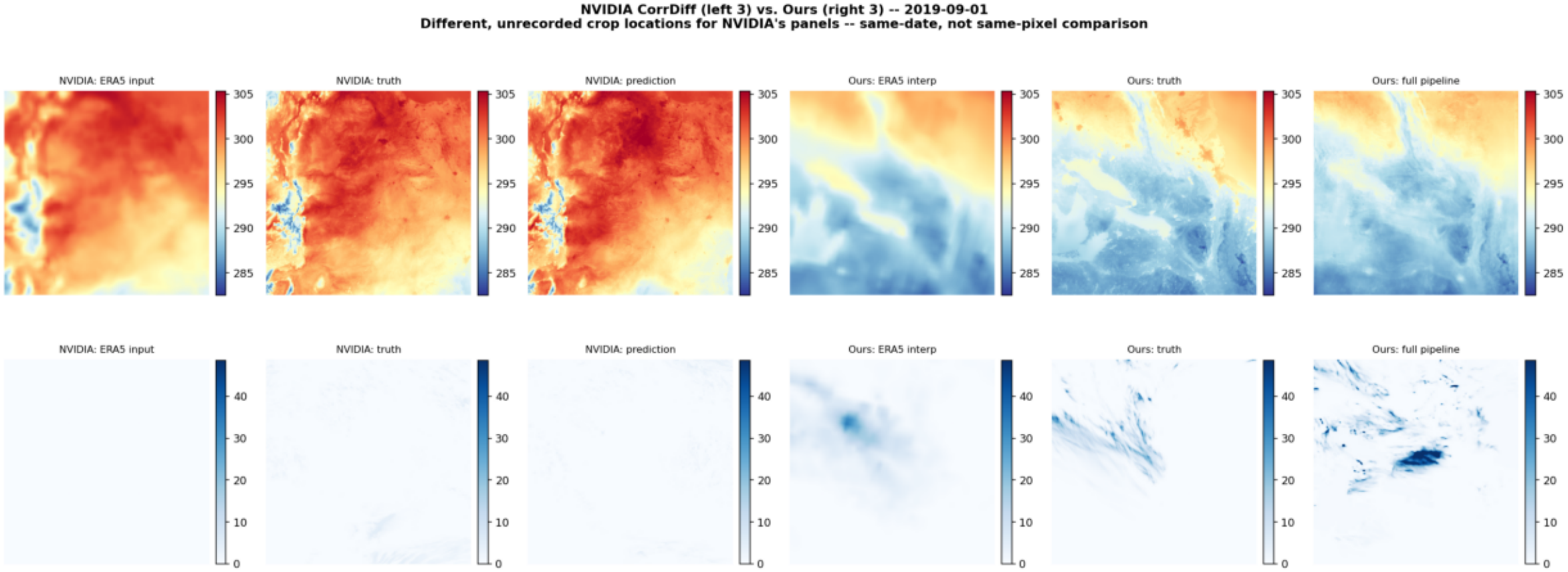
NVIDIA CorrDiff vs. ours, 2019-08-18 — same date, different (unrecorded) crops



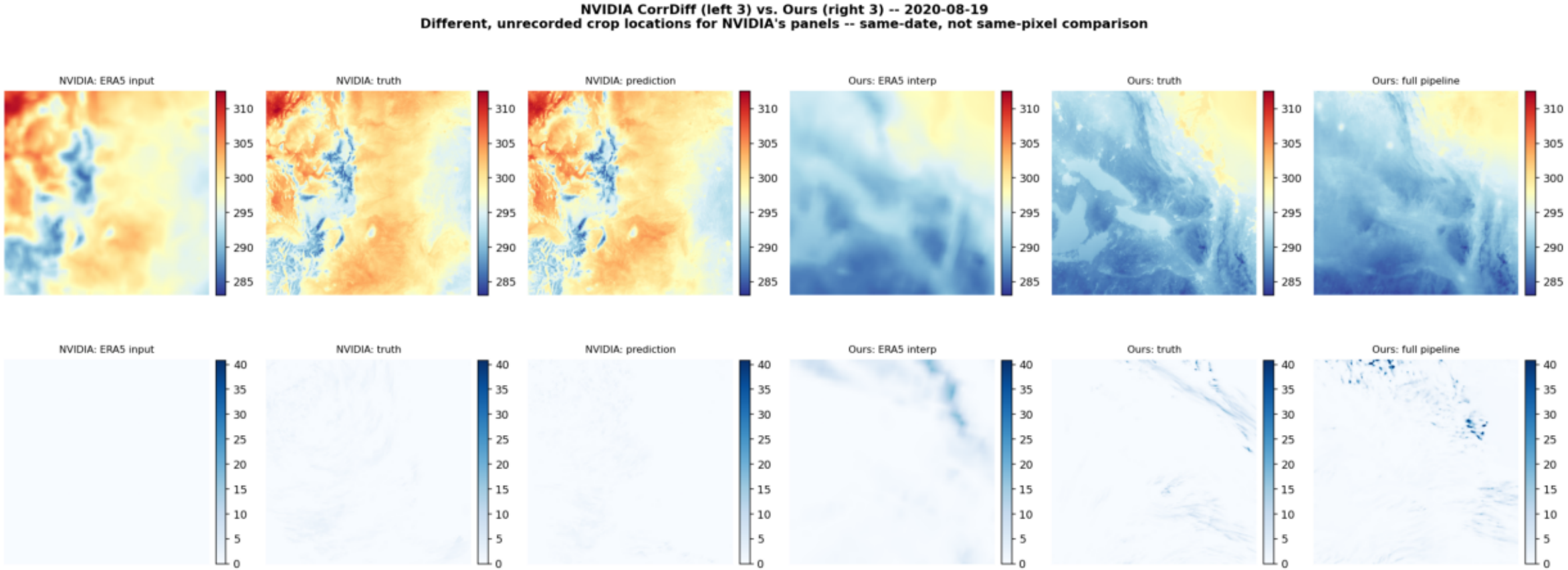
NVIDIA CorrDiff vs. ours, 2019-08-25 — same date, different (unrecorded) crops



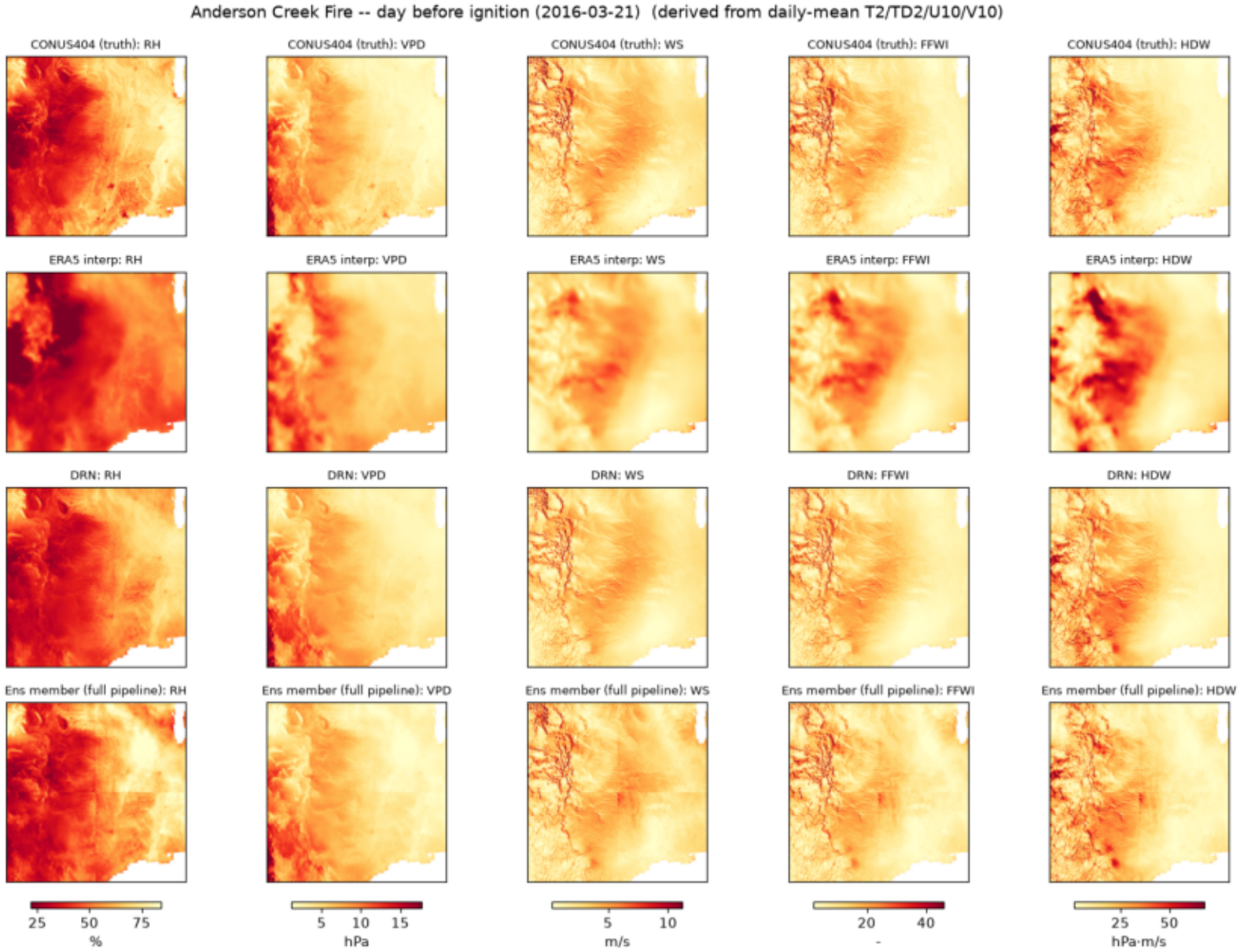
NVIDIA CorrDiff vs. ours, 2019-09-01 — same date, different (unrecorded) crops



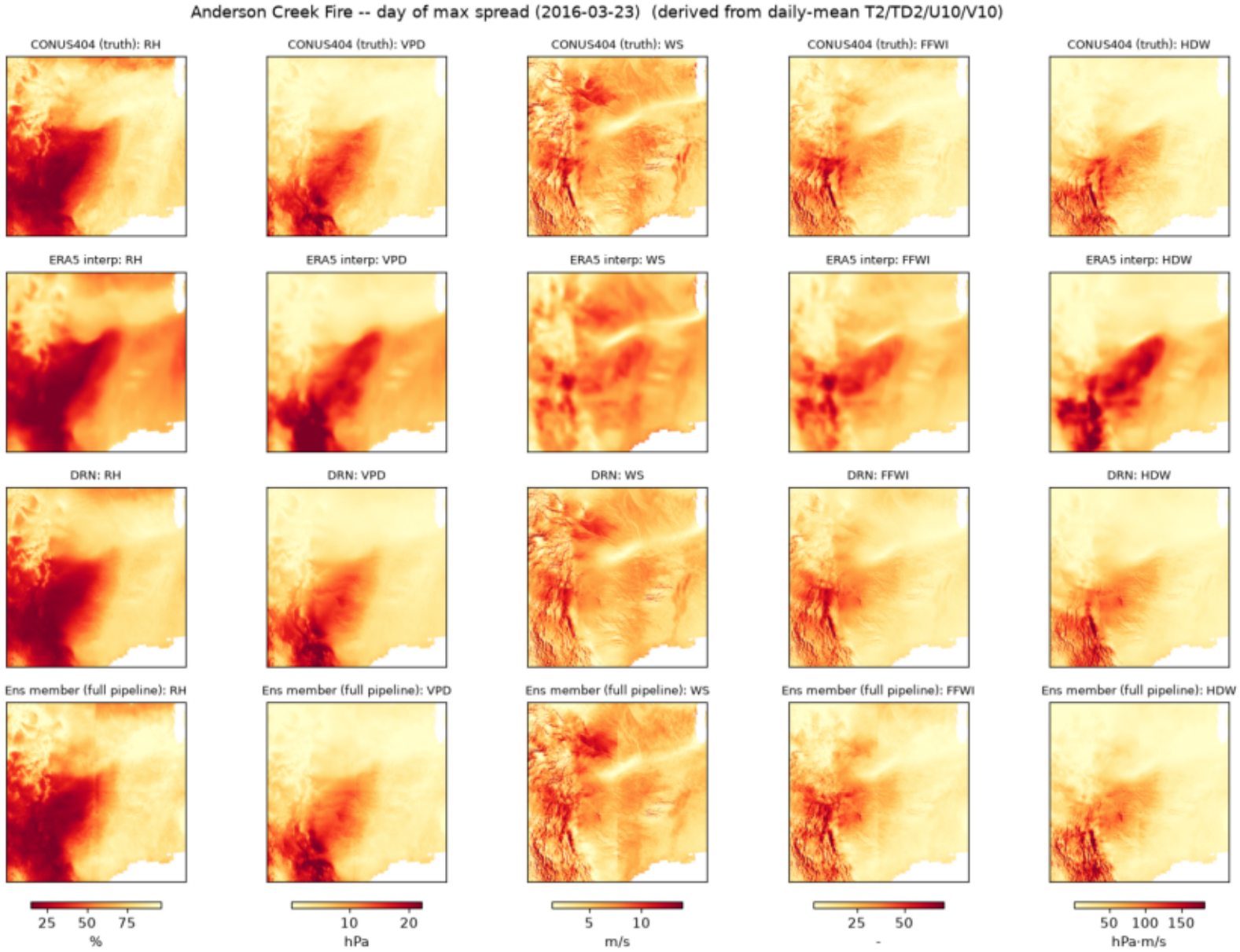
NVIDIA CorrDiff vs. ours, 2020-08-19 — same date, different (unrecorded) crops



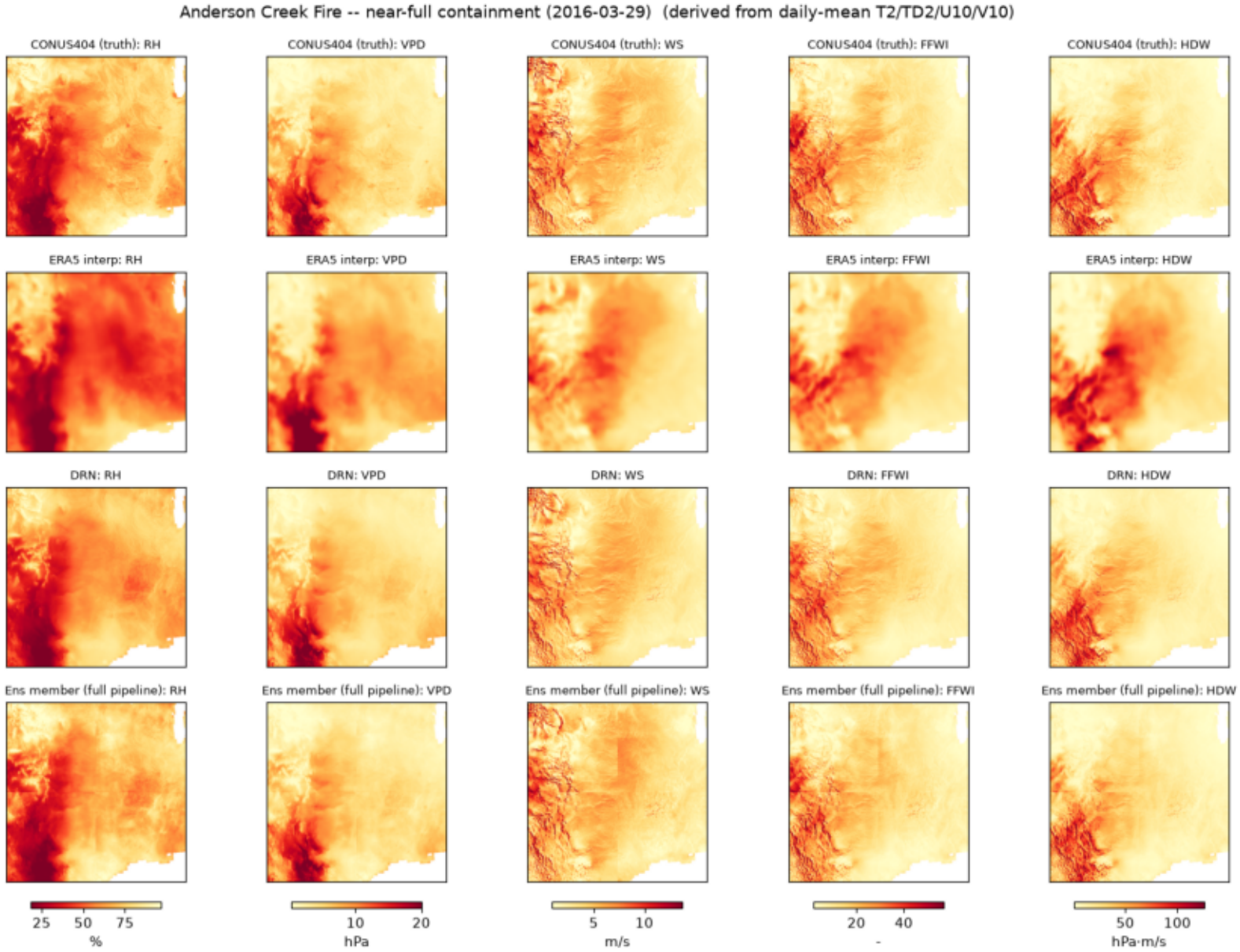
Anderson Creek Fire — day before ignition (2016-03-21)



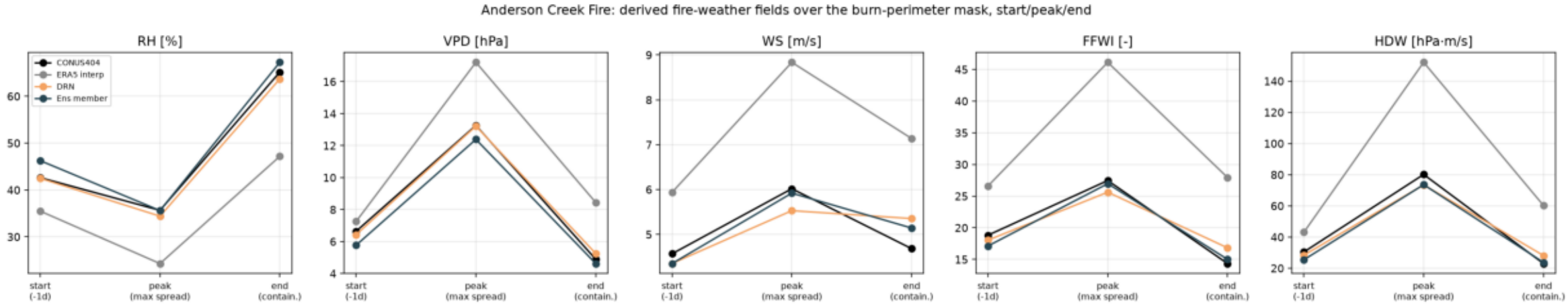
Anderson Creek Fire — day of maximum spread (2016-03-23)



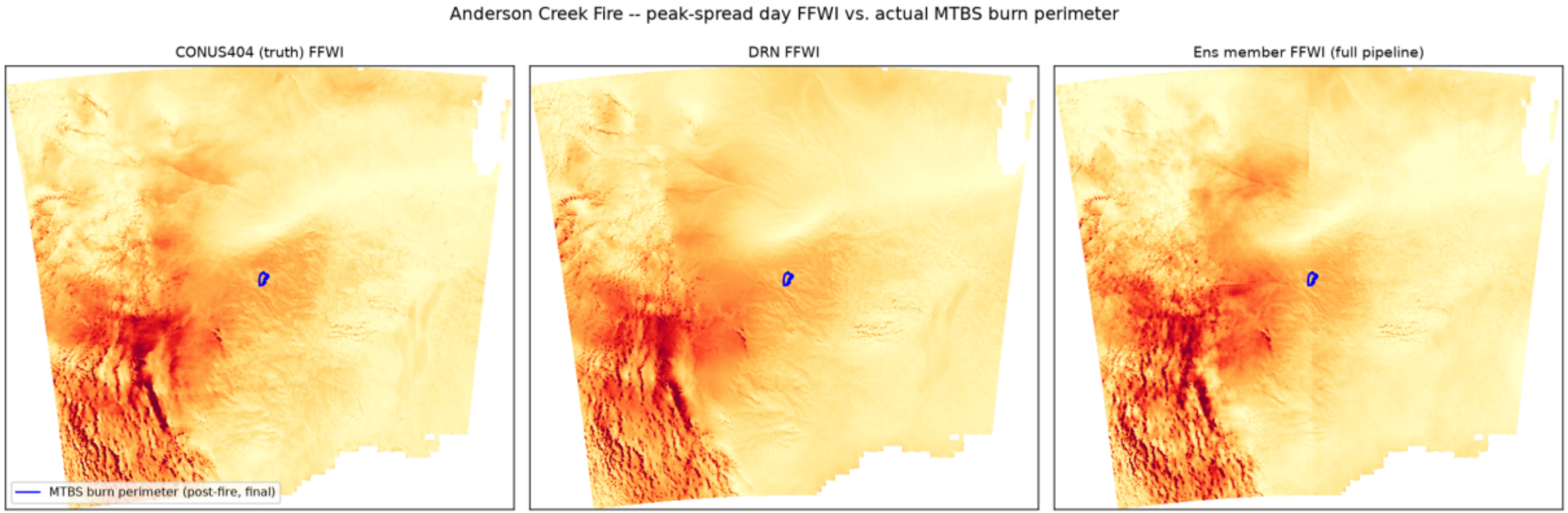
Anderson Creek Fire — near-full containment (2016-03-29)



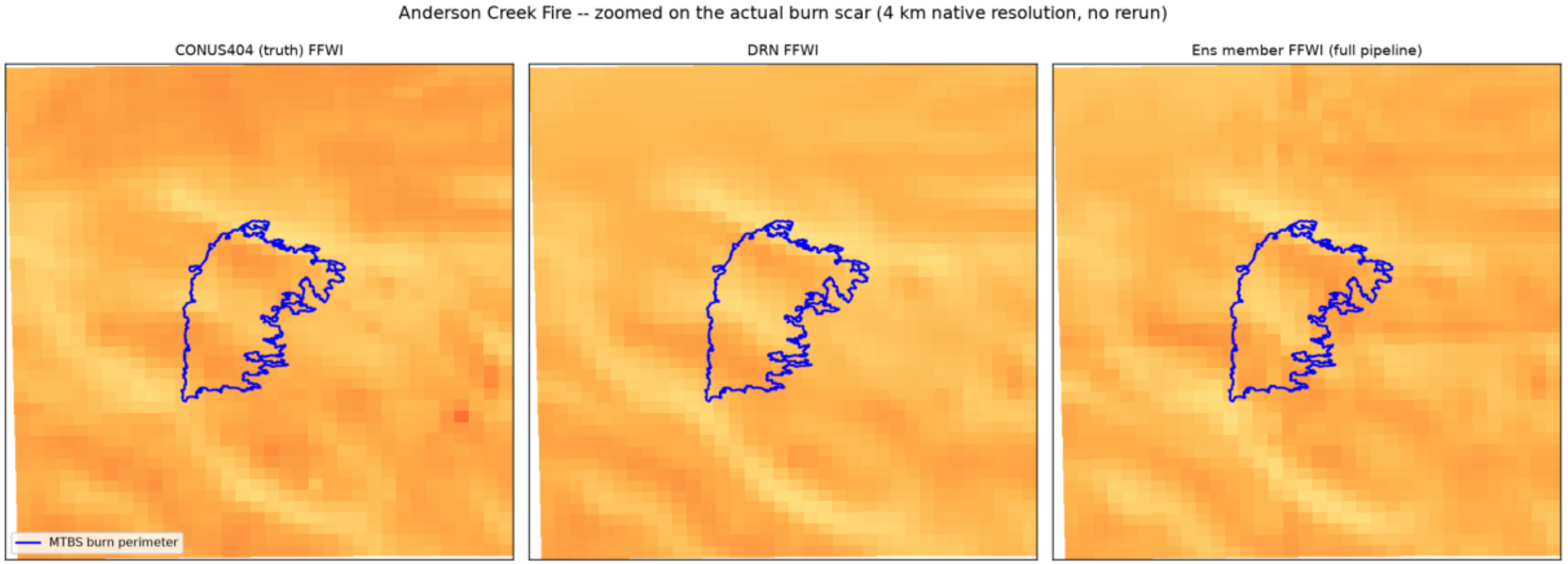
Fire-weather timeline: start → peak → containment



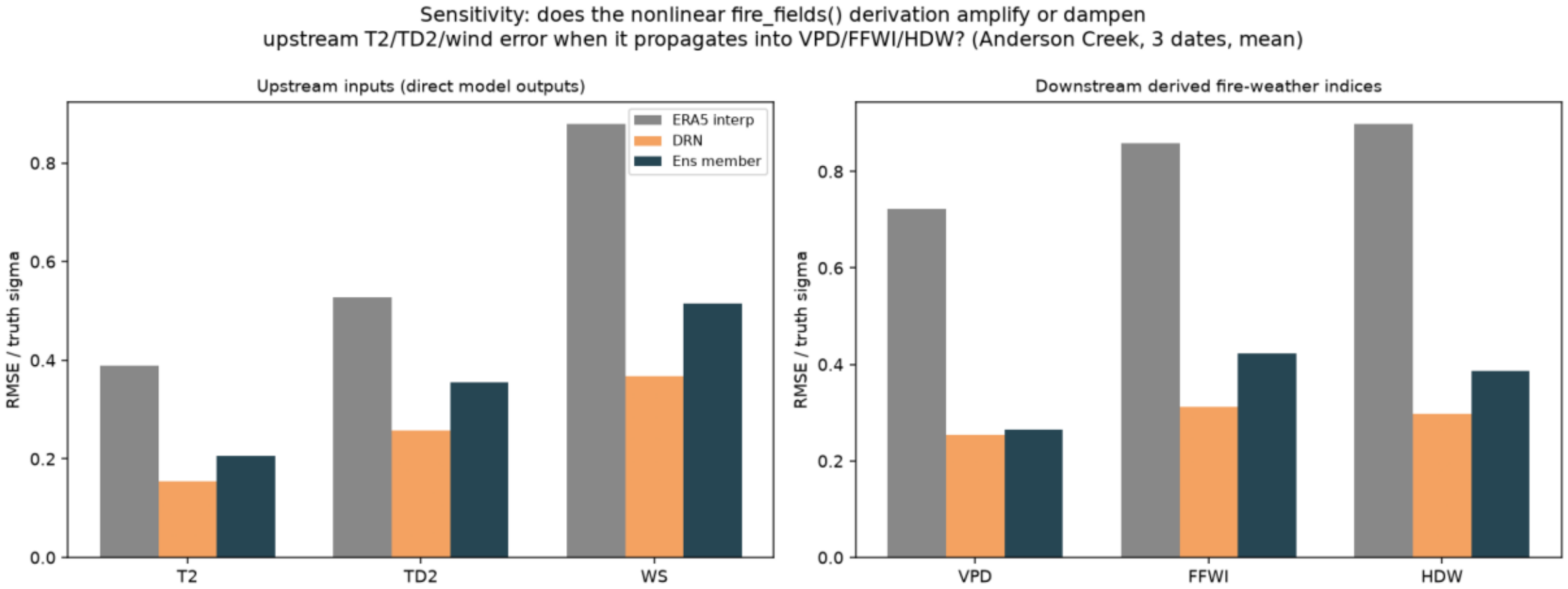
Peak-spread FFWI vs. the actual MTBS burn perimeter



Zoomed on the actual burn scar (4 km native resolution)



Sensitivity: does the fire-weather formula amplify upstream error?



Independent station check (AVK ASOS, ~46 km from the fire) vs. model

